

27/10/2021

S T Q Q S S D

① $S \rightarrow aS | bB | aB | \epsilon$
 $A \rightarrow bbB | BC | c | Ca | b | \epsilon$
 $B \rightarrow bdB | BA | CID$
 $C \rightarrow AS | aB | \epsilon | D$
 $D \rightarrow X | Y$

② Eliminar ϵ -produção:

$S \rightarrow aS | bB | aB | a | b$
 $A \rightarrow bbB | B | C | c | Ca | b$
 $B \rightarrow bdB | B | A | C | D$
 $C \rightarrow AS | aB | D$
 $D \rightarrow X | Y$

③ Eliminar produção unitários:

$S \rightarrow aS | bB | aB | a | b$
 $A \rightarrow bbB | B | A | S | aB | D | c | Ca | b$
 $B \rightarrow bdB | B | A | S | aB | D$
 $C \rightarrow A | aS | bB | aB | a | b | aB | D$
 $D \rightarrow X | Y$

④ Eliminar produção inúteis:

$S \rightarrow aS | bB | aB | a | b$
 $A \rightarrow bbB | B | A | S | aB | D | c | Ca | b$
 $B \rightarrow bdB | B | A | S | aB$
 $C \rightarrow A | aS | bB | aB | a | b | aB$

④ Chomsky:

$S \rightarrow XS | YB | XB | a | b$

$A \rightarrow Y_2 B | B | A | S | X | B | c | C | X | b$

$B \rightarrow Y_3 B | B | A | S | X | B$

$C \rightarrow A | X | S | Y | B | X | B | a | b$

$X \rightarrow a$

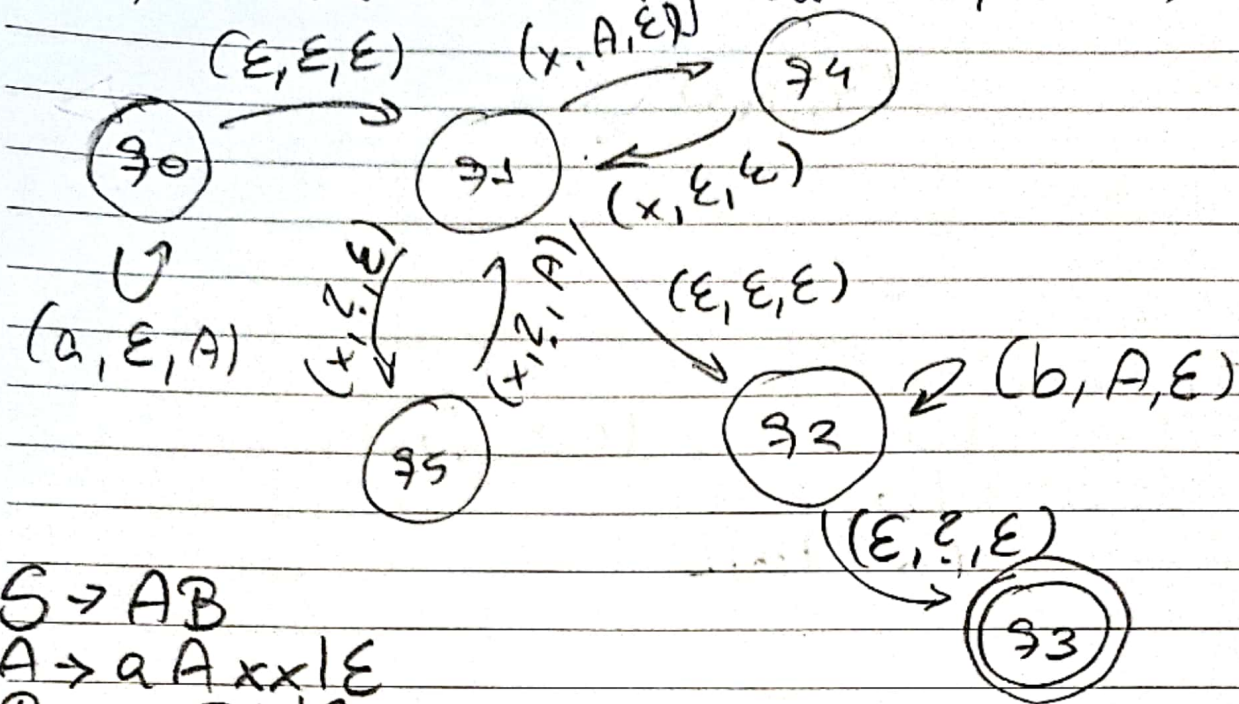
$Y \rightarrow b$

$Y_2 \rightarrow bb$

$Y_3 \rightarrow bcb$

②

$$a) L = \{a^i x^j b^k \mid j = 2i + 2k, k \geq 0\}$$

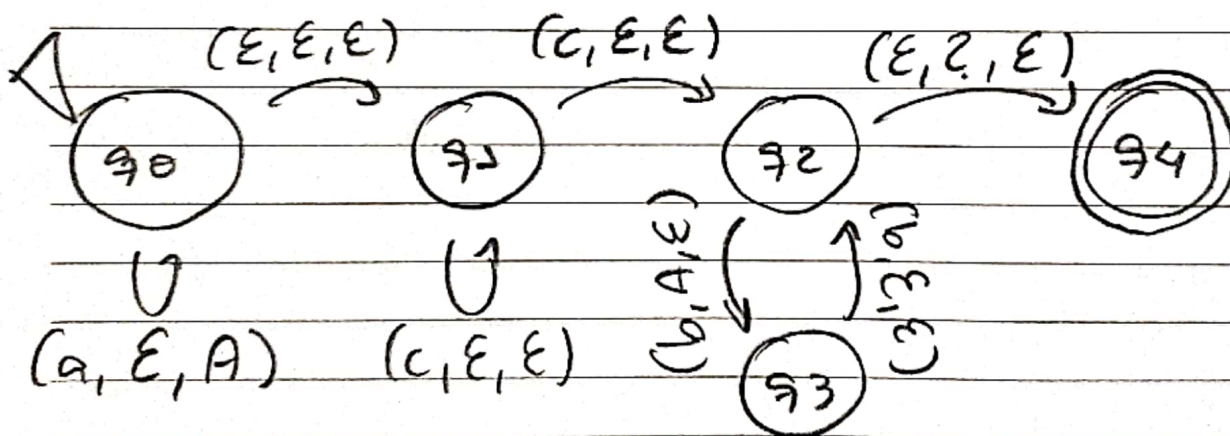


$$S \rightarrow AB$$

$$A \rightarrow aAx | \epsilon$$

$$B \rightarrow xxBb | \epsilon$$

$$b) L = \{a c^z b^{2n} \mid n \geq 0, z > 0\}$$



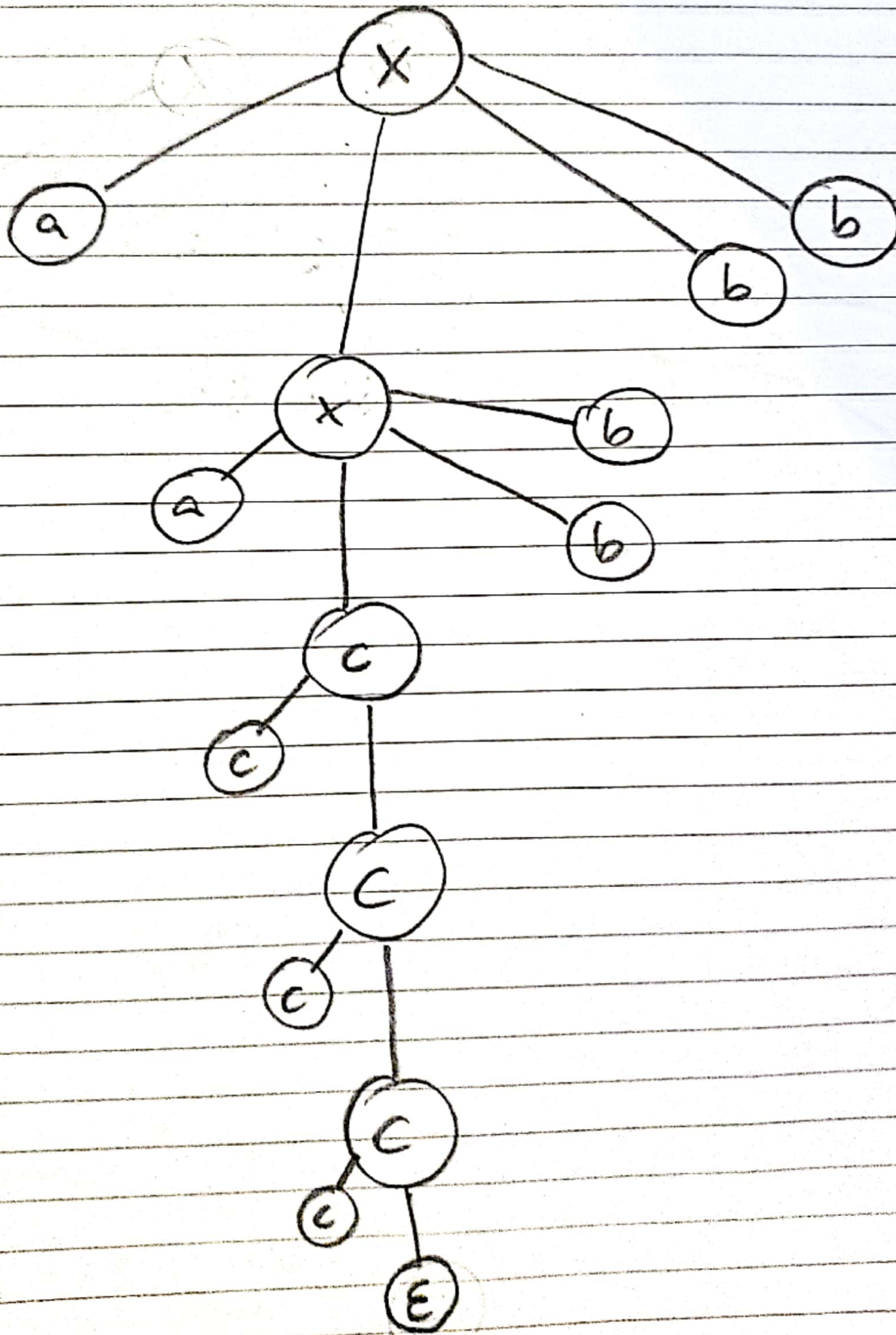
$$S \rightarrow aXbb$$

$$X \rightarrow cC | aXbb$$

$$C \rightarrow cC | \epsilon$$

③ GNC da alternativa B

palavra: ac ccc bbb b



④ Mealy:

